Homework 5
17 February 2025
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## *Question 1*

**Given:**

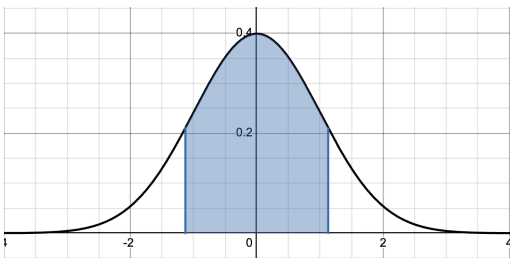
The confidence interval $\bar{x} - \frac{1.13\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + \frac{1.13\sigma}{\sqrt{n}}$

ZtableNegativeInfinity.pdf

**Find:**

The confidence level for the interval.

**Diagram:**




**Theory:**

Table ZtableNegativeInfinity.pdf has the P values of Z ≥ 0.

**Assumptions:**

Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

**Solution:**

Since we can see that $z = 1.13$

$P(Z \pm z) = P(-z \leq Z \leq z) = P(-1.13 \leq Z \leq 1.13)$

Using the Z-table we find that

$[P(Z \leq 1.13) = 0.8708] - [P(Z \leq -1.13) = 0.1209] = 0.7416$

Therefore the confidence level is **74.16%**.

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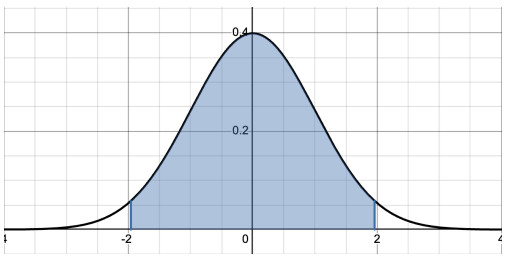
## *Question 2*

**Given:**
A sample of 91 college students handled 216 messages. A 95% confidence interval 210 to 222.

**Find:**
a) The population standard deviation.
b) The new confidence interval if the number of samples were doubled.

**Diagram:**




**Theory:**
Confidence Interval: $\bar{x} \pm Z\frac{\sigma}{\sqrt{n}}$

**Assumptions:**
Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

**Solution:**

a) 95% confidence interval yields $Z = 1.96$
Confidence interval $(X_{low}, X_{high}) = (210, 222)$
We find that $X_{high} = \bar{x} + Z\frac{\sigma}{\sqrt{n}}$,
rearranged to solve for σ $\sigma = \frac{\sqrt{n}(X_{high} - \bar{x})}{Z} = \frac{\sqrt{91}(222-216)}{1.96} = 29.18$
Therefore the standard deviation is approximately **29.18**.

b) Reversing with new sample size of 182
$2\bar{1}6 \pm 1.96\frac{29.18}{\sqrt{182}} = (211.8,\ 220.2)$
Therefore the new confidence interval is **(211.8, 220.2)**.

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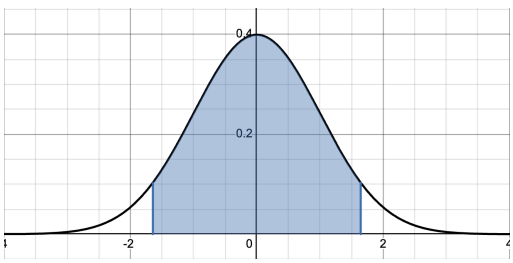
## *Question 3*

**Given:**
A 90% confidence interval to be within 2 days needs a sample size of 102.

**Find:**
The sample size to yield a 90% confidence interval to be within 1 day.

**Diagram:**




**Theory:**
Confidence interval $\bar{x} \pm E = \bar{x} \pm Z\frac{\sigma}{\sqrt{n}}$
Because margin of error is directly proportional to the square root of sample size, therefore sample size is directly proportional to the square of the margin of error, we conclude
$n_2 = n_1(\frac{E_2}{E_1})^2$

**Assumptions:**
Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

**Solution:**
$n_2 = 102(\frac{2}{1})^2 = 102(4) = 408$

There needs to be a sample size of $\boxed{\textbf{408}}$ to yield the desired interval.

996 × 1,000